

Department of Electrical & Electronic Engineering

BRAC University

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EEE309

Semiconductor Device Physics

Section: 02

Project:

Numerical and Experimental Analysis of MOSFET Using MATLAB

Group Number:(4)

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Given Parameters:

For a Silicon-based n-channel MOSFET with the following device-specific parameters:

Parameter	Value
Oxide Thickness (t_{ox})	$t_{ox} = 10 \text{ nm}$
Oxide Relative Permittivity	$\epsilon_{ox} = 3.9 \text{ (SiO}_2\text{)}$
Channel Length	$L = 1 \text{ }\mu\text{m}$
Channel Width	$W = 10 \text{ }\mu\text{m}$
Substrate Doping (p-type)	$N_a = \beta \times 10^{14} \text{ cm}^{-3}$
Flat-band Voltage	$V_{OB} = -0.9 \text{ V (n}^+\text{-poly gate)}$
Fixed Oxide Charge Density	$Q_f / q = 2 \times 10^{11} \text{ cm}^{-2}$
Source/Drain Extension Doping	$N_D = (100 - \beta) \times 10^{15} \text{ cm}^{-3}$

Task 1: Threshold Voltage and Depletion Width Analysis

a)

Calculate the following parameters for your MOSFET at zero gate bias ($V_{G_s} = 0$):

- Fermi potential (ϕ^0) of the p-type substrate
- Maximum depletion width ($x_D^{T_a^x}$) at the onset of strong inversion
- Maximum depletion charge density ($Q_D^{T_a^x}$)
- Oxide capacitance per unit area (C_{ox})
- Threshold voltage (V_{th}) using the standard MOS threshold voltage formula

b) Plot the depletion width (x_D) as a function of gate voltage (V_{G_s}) for the range $[-2\text{V to } +3\text{V}]$ using MATLAB, from accumulation through depletion to strong inversion. Mark the threshold voltage V_{th} on the plot.

c) Explain the physical changes occurring in the MOSFET channel region as gate voltage increases from accumulation to strong inversion. Discuss how threshold voltage is affected by substrate doping concentration.

Task 1

EEE 309

Project: Numerical and Experimental Analysis of MOSFET using MATLAB.

Task: 1 Threshold Voltage and Depletion Width Analysis.

Given parameters,

$$\beta = 4 \quad [\text{Group no.} - 4]$$

MOSFET Parameters:

$$\text{Oxide thickness, } (t_{ox}) = 10 \text{ nm} = 10 \times 10^{-9} \text{ m}$$

$$\text{Oxide relative permittivity, } \epsilon_{ox} = 3.9 (\text{SiO}_2)$$

$$\text{channel length, } L = 1 \mu\text{m} = 1 \times 10^{-4} \text{ m}$$

$$\text{channel width, } W = 10 \mu\text{m} = 10 \times 10^{-4} \text{ m} = 1 \times 10^{-3} \text{ m}$$

$$\begin{aligned} \text{Substrate doping (p-type), } N_a &= 1.2 \times 10^{14} \text{ cm}^{-3} \\ &= 4 \times 10^{14} \text{ cm}^{-3} \end{aligned}$$

$$\text{Flat Band voltage, } V_{FB} = -0.9 \text{ V (n}^+\text{-poly gate)}$$

$$\text{Fixed oxide charge density, } \frac{Q_f}{q} = 2 \times 10^{11} \text{ cm}^{-2}$$

$$\begin{aligned} \text{Source/Drain Extension Doping, } N_D &= (100 - \beta) \times 10^{15} \text{ cm}^{-3} \\ &= 96 \times 10^{15} \text{ cm}^{-3} \end{aligned}$$

$$N_D = 9.6 \times 10^{16} \text{ cm}^{-3}$$

$$\text{Also, } \frac{W}{L} = \frac{1 \times 10^{-3}}{1 \times 10^{-4}} = 10$$

Task 1(a)

We have to calculate,

- ① * Fermi potential ϕ^{FP}
- ② * Maximum depletion width, W_{dmax}
- ③ * Maximum depletion charge density, Q_{Dmax}
- ④ * Oxide capacitance, C_{ox}
- ⑤ * Threshold Voltage, V_{TH}

First,

given, $\epsilon_{Si} = 11.7 \epsilon_0$ [Silicon permittivity]
 $= 11.7 (8.854 \times 10^{-14})$
 $= ~~1.0359~~ 1.035918 \times 10^{-12} \text{ F/cm}$

Now,

Oxide permittivity, $\epsilon_{ox} = 3.9 \epsilon_0$
 $= 3.9 (8.854 \times 10^{-14})$
 $= 3.45306 \times 10^{-13} \text{ F/cm}$

①

So, Fermi Potential, [for p type substrate]

$$\phi^{FP} = \frac{kT}{q} \ln\left(\frac{N_A}{n_i}\right) \quad \left[\frac{kT}{q} = 0.02585 \text{ V} \right]$$

@ $T = 300 \text{ K}$

$$\text{So, } \phi_{FP} = 0.02585 \ln\left(\frac{4 \times 10^{14}}{1.5 \times 10^{10}}\right) \quad \left[\text{Given, } n_i = 1.5 \times 10^{10} \right]$$

$$= ~~0.6344 \text{ V}~~ \\ = 0.26344 \text{ V} \quad (\text{Ans.})$$

② Maximum Depletion Width,

$$\begin{aligned}
 W_{dmax} &= \sqrt{\frac{4\epsilon_{si} \phi_{FP}}{eN_A}} \\
 &= \sqrt{\frac{4(1.035918 \times 10^{-12})(0.26344)}{(1.602 \times 10^{-19})(4 \times 10^{14})}} \\
 &= 1.30519 \times 10^{-4} \text{ cm} \\
 &= 1.305 \mu\text{m} \quad (\text{Ans.})
 \end{aligned}$$

③ Maximum depletion charge density,

$$\begin{aligned}
 Q_{dmax} &= \sqrt{4e\epsilon_{si} N_A \phi_{FP}} \\
 &= \sqrt{4(1.602 \times 10^{-19})(1.035918 \times 10^{-12})(4 \times 10^{14})(0.26344)} \\
 &= 8.3636 \times 10^{-9} \text{ C/cm}^2 \\
 &\quad (\text{Ans.})
 \end{aligned}$$

④ Oxide capacitance per unit area,

$$\begin{aligned}
 \text{oxide capacitance} &= C_{ox} = \frac{\epsilon_{ox} A}{d} \\
 \text{" " per unit Area} &= \frac{C_{ox}}{A} = \frac{\epsilon_{ox}}{t_{ox}}
 \end{aligned}$$

$$\begin{aligned}
 \text{So, } \frac{C_{ox}}{A} &= \frac{\epsilon_{ox}}{t_{ox}} = \frac{3.9\epsilon_0}{10 \times 10^{-7}} = \frac{3.45306 \times 10^{-13}}{10 \times 10^{-7}} \\
 &= 3.45306 \times 10^{-7} \text{ F/cm}^2 \\
 &\quad (\text{Ans.})
 \end{aligned}$$

⑤ Threshold voltage,

As V_{FB} is given,

$$V_{TH} = V_{FB} + 2\phi_{FP} + \frac{Q_{dmax}}{\frac{C_{ox}}{A}}$$
$$= -0.9 + 2(0.26344) + \frac{8.364 \times 10^{-9}}{\frac{3.45306 \times 10^{-7}}{A}}$$

$$= -0.3489V$$

$$\approx -0.349V$$

(Ans.)

Task 1(b) From (a) $\rightarrow V_{TH} = -0.349V$

$$W_{Dmax} = 1.305 \mu m = 1.305 \times 10^{-4} cm$$

Gate voltage range $-2V \leq V_{GS} \leq 3V$

We know,

$$W_D = \sqrt{\frac{2\epsilon_{si}\psi_s}{eN_a}} \quad [\because \text{depletion width}]$$

To determine the surface potential, (ψ_s) we used this relationship

$$V_{GS} - V_{FB} = \psi_s + \frac{\sqrt{2e\epsilon_{si}N_a\psi_s}}{C_{ox}}$$

For Accumulation Region,

$$\text{where, } V_{GS} \leq V_{FB}$$

$$\text{as, } V_{FB} = -0.9V$$

So, the MOS structure operates in the accumulation region. In this region, holes accumulate near the surface, so, $W_D = 0$ [Depletion width]

For Depletion Region; where, $V_{FB} < V_{GS} < V_{TH}$

$$\text{or, } -0.9 < V_{GS} < -0.349V$$

Here, holes are repelled from the surface, which forms a depletion region.

$$\text{so, } W_D = \sqrt{\frac{2\epsilon_{si}\psi_s}{eN_a}}$$

when $V_{GS} \uparrow$, W_D also $\uparrow$

For Inversion Region; $V_{GS} \geq V_{TH}$

where, $V_{TH} \geq -0.349V$

The MOSFET enters into inversion, the depletion width (W_D) reaches maximum value,

$$\text{So, } W_D = W_{D\max} = 1.305 \mu\text{m}$$

When, $V_{GS} \uparrow$, The W_D remains constant.

Code for Task 1(b):

```
clear;  
close all;  
clc;
```

% Task 1(b) Group 4

```
vg = -2:0.001:3;  
q = 1.6e-19;  
e0 = 8.854e-14;  
es = 11.7*e0;  
cox = 3.453e-7;  
na = 4e14;  
vfb = -0.9;  
vth = -0.349;%from task 1a  
xdmax = 1.305e-4;%also from task 1a
```

```

xd = zeros(size(vg));
a = sqrt(2*q*es*na)/cox;

for i = 1:length(vg)

    if vg(i) <= vfb
        xd(i) = 0;
    elseif vg(i) < vth
        v = vg(i) - vfb;
        y = (-a + sqrt(a^2 + 4*v))/2;
        ps = y^2;
        xd(i) = sqrt((2*es*ps)/(q*na));
    else
        xd(i) = xmax;
    end

end

% Convert cm to um
xd = xd*1e4;

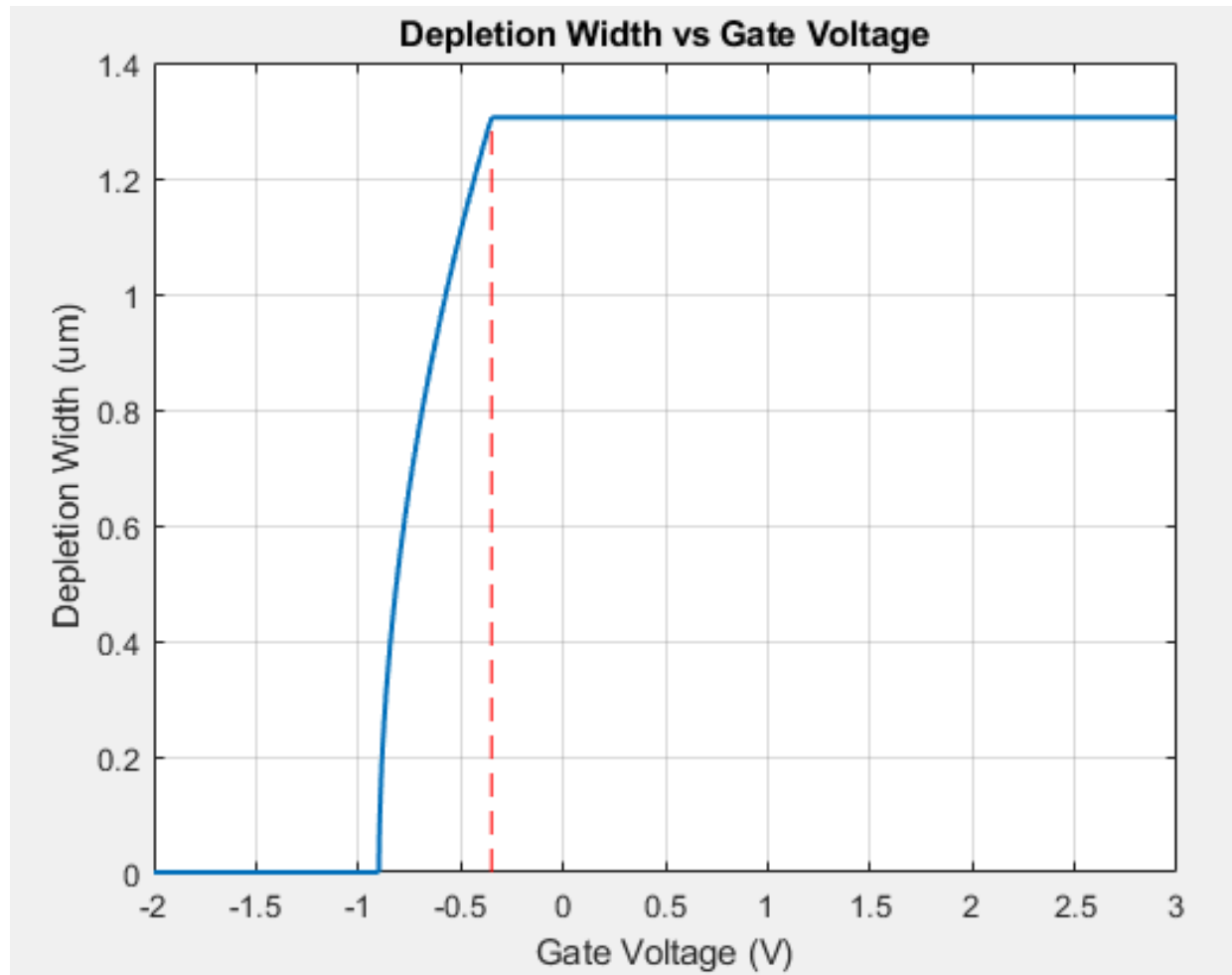
figure

plot(vg,xd,'LineWidth',1.5)
grid on
hold on
plot([vth vth],[0 xmax*1e4],'--r')
xlabel('Gate Voltage (V)')
ylabel('Depletion Width (um)')

title('Depletion Width vs Gate Voltage')

```

Output:



Fig; Plot of Depletion width (W_D) Vs Gate voltage (V_{GS})

Task 1(c) :

As the gate voltage increases, the MOSFET goes through accumulation, depletion, and strong inversion.

At a negative or low gate voltage, the MOSFET is in the accumulation region. Since the substrate is p-type, holes are attracted towards the semiconductor and accumulate near the oxide part. $[V_{GS} \leq V_{FB}]$

As the gate voltage becomes more positive, the holes are repelled away from the surface. This creates a depletion region. The removal of holes leave behind negative(-) charged acceptors. As the gate voltage increases further, the depletion region gets wider $[V_{FB} < V_{GS} < V_{TH}]$

When the gate voltage reaches Threshold Voltage, electrons begin to accumulate near the semiconductor surface. This creates an inversion. The negative carriers create a channel between the n type terminals. $[V_{GS} = V_{TH}]$

When $V_{GS} > V_{TH}$ or, when the gate voltage is greater than the Threshold voltage, the electrons keep on accumulating but the depletion channel

width remain the same.

Effect of Substrate Doping on V_{TH} :

If the substrate doping concentration increases, more ionized acceptor atoms are present in the MOSFET substrate. And a greater Threshold voltage (V_{TH}) is required to create inversion.

So, when N_a is increased, V_{TH} is increased,

so, $N_a \uparrow$; $V_{TH} \uparrow$.

(Ans.)

Task 2

Task 2: MOSFET I-V Characteristics – Transfer and Output Curves

a)

Using the standard long-channel MOSFET drain current model, compute the drain current (I_D) for each bias point listed below. The three operating regions are defined as follows:

Region	Drain Current Equation	Validity Condition
Cut-off	$I_D = 0$	$V_{GS} < V_{TH}$
Linear	$I_D = \mu_{eff} \cdot C_{ox} \cdot (W/L) \cdot [(V_{GS} - V_{TH}) \cdot V_{DS} - V_{DS}^2 / 2]$	$V_{DS} < V_{GS} - V_{TH}$
Saturation	$I_D = (\mu_{eff} \cdot C_{ox} \cdot W) / (2L) \cdot (V_{GS} - V_{TH})^2$	$V_{DS} \geq V_{GS} - V_{TH}$

Evaluate I_D at every combination of the following bias voltages:

- $V_{GS} = V_{TH}, V_{TH} + 0.5 \text{ V}, V_{TH} + 1.0 \text{ V}, V_{TH} + 1.5 \text{ V}, V_{TH} + 2.0 \text{ V}$
- $V_{DS} = 0 \text{ V}, 0.5 \text{ V}, 1.0 \text{ V}, 1.5 \text{ V}, 2.0 \text{ V}, 2.5 \text{ V}, 3.0 \text{ V}$

Tabulate the calculated I_D values and identify the operating region (Cut-off / Linear / Saturation) for each bias point. Additionally, derive the transconductance $g_m = \partial I_D / \partial V_{GS}$ at the bias point $V_{DS} = 2 \text{ V}$, $V_{GS} = V_{TH} + 1 \text{ V}$.

b) Generate the following two plots in MATLAB:

- Output Characteristics** — I_D vs. V_{DS} for four gate overdrive levels: $V_{GS} = V_{TH} + \{0.5, 1.0, 1.5, 2.0\} \text{ V}$, swept over $V_{DS} \in [0, 3] \text{ V}$. Mark the linear-saturation boundary ($V_{DS,sat} = V_{GS} - V_{TH}$) with a dashed locus on the plot.
- Transfer Characteristics** — I_D vs. V_{GS} at a fixed $V_{DS} = 2 \text{ V}$, swept over $V_{GS} \in [0, V_{TH} + 2 \text{ V}]$. On a second subplot, plot $\sqrt{I_D}$ vs. V_{GS} ; extrapolate the linear portion to the V_{GS} -axis to graphically extract V_{TH} .

Part A

Task-2

Given,

oxide thickness $t_{ox} = 10 \text{ nm}$

Oxide Relative Permittivity $\epsilon_o = 3.9$

width $W = 10 \mu\text{m}$.

Length $L = 1 \mu\text{m}$

Effective e^- mobility

$\mu_{eff} = 450 \text{ cm}^2/\text{Vs}$

from, task 1, $V_{TH} = -0.3489 \text{ V}$

Here,

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$$

$$= \frac{3.9 \times 8.854 \times 10^{-14}}{10^{-6}} \left[10 \text{ nm} = 10^{-6} \text{ cm} \right]$$

$$= 3.45 \times 10^{-7} \text{ F/cm}^2$$

again,

$$K = \mu_{eff} \cdot C_{ox} \cdot \frac{W}{L}$$

$$= (450) (3.45 \times 10^{-7}) \left(\frac{10}{1} \right)$$

$$= 1.5 \times 10^{-3} \text{ A/V}^2$$

$$= 1.55 \text{ mA/V}^2$$

now, V_{GS}

$$V_{GS1} = V_{TH}$$

$$= -0.3489 \text{ V}$$

$$V_{GS2} = V_{TH} + 0.5$$

$$= -0.3489 + 0.5 = 0.1511 \text{ V}$$

$$V_{GS3} = V_{TH} + 1$$

$$= -0.3489 + 1 = 0.6511 \text{ V}$$

$$V_{GS4} = V_{TH} + 1.5$$

$$= -0.3489 + 1.5 = 1.1511 \text{ V}$$

$$V_{GS5} = V_{TH} + 2.0$$

$$= -0.3489 + 2 = 1.6511 \text{ V}$$

again,

$$V_{GS} = 0.00571, 0.005$$

now,

case-1

$$\text{At } V_{GS1} = -0.3489 \text{ V}$$

$$\therefore V_{DS} = V_{GS1} - V_{TH} = -0.3489 + 0.3489 = 0 \text{ V}$$

$$\therefore I_D = \frac{K}{2} (0)^2$$

$$= 0 \text{ V}$$

The channel conductivity is zero

So, $I_D = 0_{mA}$ for $V_{DS} \{0V, 0.5V, 1V, 1.5V, 2V, 2.5V, 3V\}$

case-2

At $V_{GS2} = 0.1511V$,

$$V_{DS, sat} = V_{GS2} - V_{TH}$$

$$= 0.1511 + 0.3489$$

$$= 0.5V$$

$$I_D = \frac{K}{2} (0.5)^2$$

$$= \frac{1.5 \times 10^{-3}}{2} (0.5)^2$$

$$= 1.9 \times 10^{-4} A = 0.19 mA$$

$\therefore I_D = 0.19 mA$ for $V_{DS} \{1V, 1.5V, 2V, 2.5V, 3V\}$

case-3

At $V_{GS3} = 0.6511V$

$$V_{DS, sat} = V_{GS3} - V_{TH} = 0.6511 + 0.3489$$

$$= 1.0V$$

At, $V_{DS} = 0.5V$

Since, $0.5 < 1V$ $\therefore$ It is in linear region.

$$I_D = K [(V_{GS} - V_{TH})^2]$$

$$I_D = K \left[(V_{DS, sat}) (V_{DS}) - \frac{V_{DS}^2}{2} \right]$$

$$= (1.5 \times 10^{-3}) \left[(0.5)(1) - \frac{0.5^2}{2} \right]$$

$$= 0.58 \text{ mA}$$

At $V_{DS} = 1V$, it is a saturation region.

$$I_D = \frac{K}{2} (1)^2$$

$$= \left(\frac{1.5 \times 10^{-3}}{2} \right) (1)^2$$

$$= 0.77 \text{ mA}$$

So, $I_D = 0.77 \text{ mA}$ for $V_{DS} \in \{1.5V, 2V, 2.5V, 3V\}$

Case-4

$$\text{At } V_{GS4} = 1.1511V$$

$$V_{DS, sat} = V_{GS4} - V_{TH}$$

$$= 1.1511 + 0.3489$$

$$= 1.5V$$

At $V_{DS} = 0$, it is a linear region

$$I_D = 0 \text{ mA}$$

At, $V_{DS} = 0.5V$; It is a linear region.

$$I_D = K \left[\left(\frac{V_{DS}}{V_{DS,sat}} \right) (V_{DS}) - \frac{V_{DS}^2}{2} \right]$$

$$= K \left[(1.5)(0.5) - \frac{(0.5)^2}{2} \right]$$

$$= 0.97 \text{ mA}$$

At, $V_{DS} = 1V$; It is a ~~linear~~ ^{linear} saturation region.

$$I_D = K \left[(1.5)(1) - \frac{1^2}{2} \right]$$

$$= (1.5 \times 10^{-3}) \left[(1.5)(1) - \frac{1^2}{2} \right]$$

$$= 1.55 \text{ mA}$$

At, $V_{DS} = 1.5V$; It is a saturation region.

$$I_D = \frac{K}{2} (1.5)^2$$

$$= \left(\frac{1.5 \times 10^{-3}}{2} \right) (1.5)^2$$

$$= 1.75 \text{ mA}$$

So, $I_D = 1.75 \text{ mA}$ for, $V_{DS} \{ 2V, 2.5V, 3V \}$.

case-5

At, $V_{GS} = 1.6511 \text{ V}$.

$$V_{DS, \text{sat}} = V_{GS} - V_{TH} = (1.6511 - 0.3489) = 2 \text{ V}$$

At, $V_{DS} = 0 \text{ V}$, I_D is linear region.

$$I_D = 0 \text{ mA}$$

At, $V_{DS} = 0.5 \text{ V}$, I_D is a linear region.

$$\begin{aligned} I_D &= k \left[(2)(0.5) - \frac{(0.5)^2}{2} \right] \\ &= (1.5 \times 10^{-3}) \left[(2)(0.5) - \frac{(0.5)^2}{2} \right] \\ &= 1.36 \text{ mA} \end{aligned}$$

At, $V_{DS} = 1 \text{ V}$, I_D is a linear region.

$$\begin{aligned} I_D &= k \left[(2)(1) - \frac{1^2}{2} \right] \\ &= (1.5 \times 10^{-3}) \left[(2)(1) - \frac{1^2}{2} \right] \\ &= 2.33 \text{ mA} \end{aligned}$$

At, $V_{DS} = 1.5 \text{ V}$, I_D is a linear region.

$$\begin{aligned} I_D &= k \left[(2)(1.5) - \frac{1.5^2}{2} \right] \\ &= 2.91 \text{ mA} \end{aligned}$$

At, $V_{DS} = 2V$, It is a saturation region,

$$I_D = \frac{K}{2} (2)^2$$

$$= \left(\frac{1.5 \times 10^{-3}}{2} \right) (2)^2$$

$$= 3.11 \text{ mA}$$

So; $I_D = 3.11 \text{ mA}$ for $V_{DS} \{ 2.5V, 3V \}$

I_D and operating region Table,

cut-off - C
Linear - L
saturation - S

$V_{GS} \backslash V_{DS}$	0V	0.5V	1V	1.5V	2V	2.5V	3V
-0.3489	0 C	0 C	0 C	0 C	0 C	0 C	0 C
0.1511	0 L	0.19 S	0.19 S	0.19 S	0.19 S	0.19 S	0.19 S
0.6511	0 L	0.58 L	0.77 S	0.77 S	0.77 S	0.77 S	0.77 S
1.1511	0 L	0.97 L	1.55 L	1.75 S	1.75 S	1.75 S	1.75 S
1.6511	0 L	1.36 L	2.33 L	2.91 L	3.11 S	3.11 S	3.11 S

Here given, $V_{DS} = 2V$.

$$V_{GS} = V_{TH} + 1 \\ = -0.3489 + 1 = 0.6511 V.$$

$$V_{DS, sat} = V_{GS} - V_{TH} = 0.6511 + 0.3489. \\ = 1 V$$

So,

$$V_{DS} = 2V, \quad V_{DS, sat} = 1V,$$

$$\therefore V_{DS} \geq V_{DS, sat}$$

It is in the saturation region.

$$\therefore I_D = \frac{K}{2} (V_{GS} - V_{TH})^2$$

$$\therefore g_m = \frac{\partial I_D}{\partial V_{GS}}$$

$$\therefore g_m = K(V_{GS} - V_{TH}) \\ = (1.5 \times 10^{-3}) (1) \\ = 1.5 \times 10^{-3} S = 1.55 mS \quad \underline{\text{Ans}}$$

$$\text{now, } I_D = \frac{K}{2} (1)^2 \\ = 0.77 mA \quad \underline{\text{Ans}}$$

Part b

Matlab Code

```
clc;
Clear all;
close all;
% Parameters
Vth = -0.3489;
Kn = 0.0015539;
%% Plot 1: Output Characteristics (ID vs VDS)
Vds = 0:0.02:3;
overdrives = [0.5, 1.0, 1.5, 2.0];
figure(1); hold on;
for Vov = overdrives
    Id = zeros(size(Vds));
    for i = 1:length(Vds)
        if Vds(i) < Vov
            Id(i) = Kn * (Vov * Vds(i) - 0.5 * Vds(i)^2); % Linear
        else
            Id(i) = 0.5 * Kn * (Vov^2); % Saturation
        end
    end
    plot(Vds, Id * 1000, 'LineWidth', 1.5);
end
xlabel('V_{DS} (V)');
ylabel('I_D (mA)');
title('Task 2(b): Output Characteristics');
grid on;
hold off;
%% Plot 2: Transfer Characteristics (ID vs VGS)
Vgs = linspace(-0.5, 1.8, 200);
Vds_fix = 2.0;
Id_tr = zeros(size(Vgs));
for i = 1:length(Vgs)
    Vov = Vgs(i) - Vth;
    if Vov > 0
        if Vds_fix < Vov
```

```

        Id_tr(i) = Kn * (Vov * Vds_fix - 0.5 * Vds_fix^2);
    else
        Id_tr(i) = 0.5 * Kn * (Vov^2);
    end
end
end
end
figure(2);
plot(Vgs, Id_tr * 1000, 'b', 'LineWidth', 1.5);
xlabel('V_{GS} (V)'); ylabel('I_D (mA)');
title('Task 2(b): Transfer Characteristic (V_{DS} = 2V)');
grid on;

```

Output

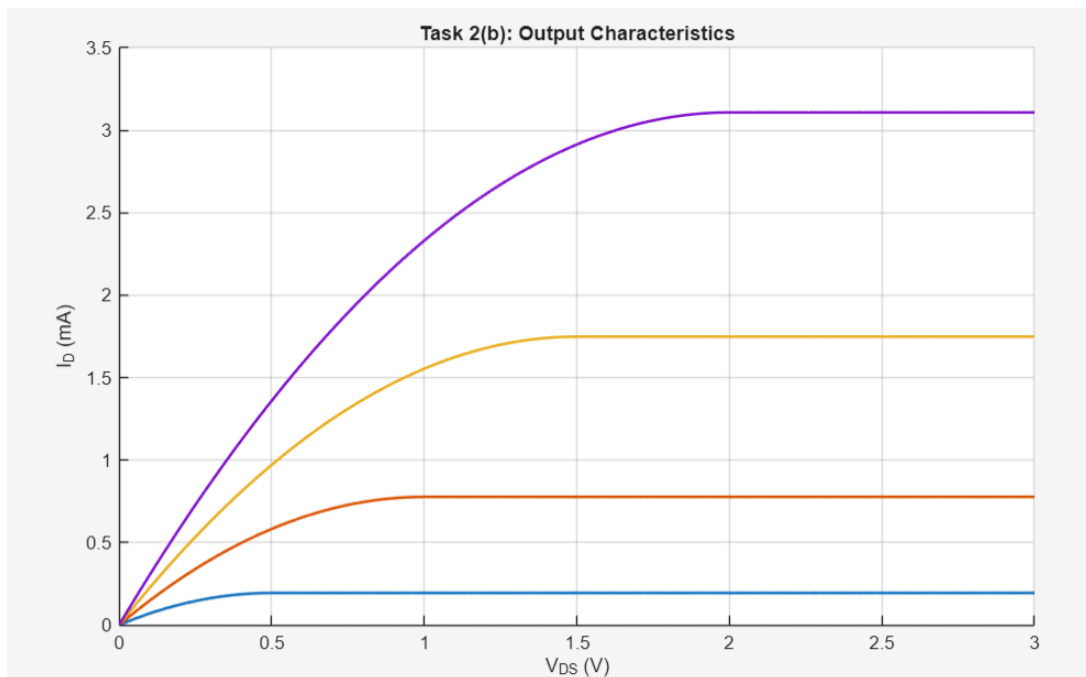


Fig: MOSFET Output Characteristics (I vs V_{DS}) for $V_{TH} = -0.3489$ V at various gate bias levels

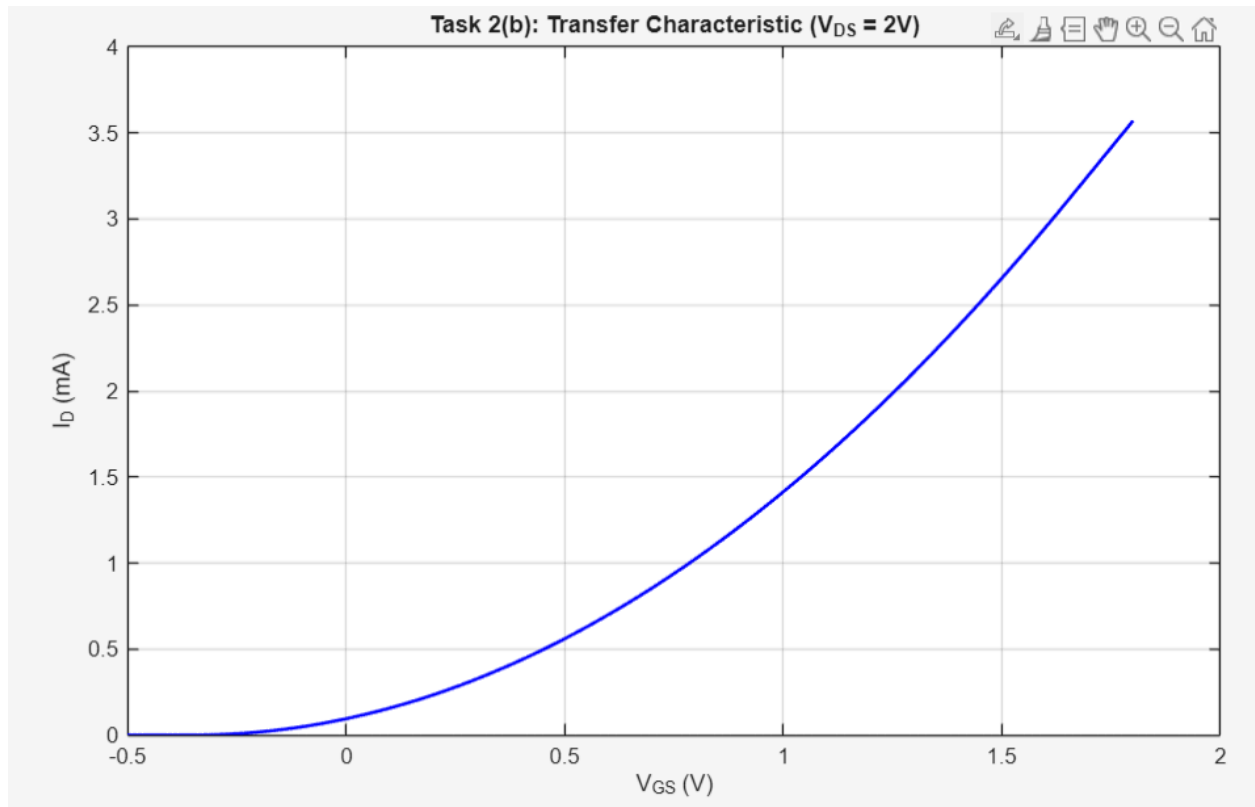


Fig: MOSFET Transfer Characteristic (I_D vs V_{GS}) at a fixed drain voltage ($V_{DS} = 2.0V$), showing turn-on behavior at $V_{TH} = -0.3489V$.

Task 3

Task 3: Effect of Device Parameters on MOSFET Characteristics

a)

Re-run the output characteristic simulation from Task 2(b) using 3 different substrate doping levels at $V_{D_s} = [0 \text{ to } 3\text{V}]$, $V_{G_s} = V_{HI} + 1\text{V}$ for each case:

- Case i: $N_a = \beta \times 10^{14} \text{ cm}^{-3}$ (original)
- Case ii: $N_a = \beta \times 10^{15} \text{ cm}^{-3}$ (10× higher doping)
- Case iii: $N_a = \beta \times 10^{13} \text{ cm}^{-3}$ (10× lower doping)

b) Plot the $I_D - V_{D_s}$ output curves for all three cases on a single graph. Recalculate V_{HI} for each case. Compare the three curves and discuss their differences in terms of saturation current, threshold voltage shift, and operating region boundaries.

c) Analyze how and why higher substrate doping concentration increases the threshold voltage, reduces the channel charge, and affects the MOSFET transconductance and on-current. Discuss implications for digital circuit design (noise margin, switching speed).

Task 03:

(a) For Group 4, $\beta = 4$.

Case I: (original)

$$N_A = \beta \times 10^{14} = 4 \times 10^{14} \text{ cm}^{-3}$$

Case II: 10 X Higher Doping

$$N_A = \beta \times 10^{15} = 4 \times 10^{15} \text{ cm}^{-3}$$

Case III: 10 X Lower Doping

$$N_A = \beta \times 10^{13} = 4 \times 10^{13} \text{ cm}^{-3}$$

$$\begin{aligned} C_{ox} &= \frac{\epsilon_{ox}}{t_{ox}} = \frac{3.9 \epsilon_0}{t_{ox}} \\ &= \frac{3.9 (8.854 \times 10^{-14})}{10^{-6}} \\ &= 3.453 \times 10^{-7} \text{ F/cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Si} \rightarrow \epsilon_{si} &= 11.7 \epsilon_0 \\ &= 11.7 (8.854 \times 10^{-14}) \\ &= 1.036 \times 10^{-12} \text{ F/cm} \end{aligned}$$

Case I: (original Doping)

$$N_A = 4 \times 10^{14} \text{ cm}^{-3}$$

$$\begin{aligned} \text{Fermi potential, } \phi_F &= \frac{kT}{q} \ln\left(\frac{N_A}{n_i}\right) \\ &= (0.0259) \ln\left(\frac{4 \times 10^{14}}{1.5 \times 10^{10}}\right) \\ &= 0.2634 \text{ V} \end{aligned}$$

$$\begin{aligned} \text{Maximum depletion width, } x_{dT} &= \sqrt{\frac{4 \epsilon_{si} \phi_F}{q N_A}} \\ &= \sqrt{\frac{4 \times (1.036 \times 10^{-12}) (0.2634)}{(1.602 \times 10^{-19}) (4 \times 10^{14})}} \\ &= 1.305 \times 10^{-4} \text{ cm} \\ &= 1.305 \mu\text{m} \end{aligned}$$

Given that,

$$t_{ox} = 10 \text{ nm} = 10^{-6} \text{ cm}$$

$$\epsilon_{ox} = 3.9 \epsilon_0$$

$$L = 1 \mu\text{m}$$

$$W = 10 \mu\text{m}$$

$$V_{FB} = -0.9 \text{ V}$$

$$n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$$

$$\mu_n = 1350 \text{ cm}^2/\text{Vs}$$

$$T = 300 \text{ K}$$

$$q = 1.602 \times 10^{-19} \text{ C}$$

$$\epsilon_0 = 8.854 \times 10^{-14} \text{ F/cm}$$

Maximum depletion charge,

$$\begin{aligned} |Q_{D,max}| &= q N_A x_{dT} \\ &= (1.602 \times 10^{-19}) \times (4 \times 10^{14}) \times (4.570 \times 10^{-5}) \\ &\quad (1.305 \times 10^{-4}) \\ &= 8.364 \times 10^{-9} \text{ C/cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Threshold Voltage, } V_{Th} &= V_{FB} + 2\phi_F + \frac{|Q_{D,max}|}{C_{ox}} \\ &= -0.9 + 2(0.2634) + \frac{8.364 \times 10^{-9}}{3.453 \times 10^{-7}} \\ &= -0.3490 \text{ V} \end{aligned}$$

$$\begin{aligned} \text{Gate voltage, } V_{GS1} &= V_{Th} + 1 \\ &= -0.3490 + 1 = 0.6510 \text{ V} \end{aligned}$$

Case II: (10x Higher Doping)

$$N_A = 4 \times 10^{15} \text{ cm}^{-3}$$

$$\begin{aligned} \phi_F &= \frac{KT}{q} \ln \left(\frac{N_A}{n_i} \right) = (0.0259) \ln \left(\frac{4 \times 10^{15}}{1.5 \times 10^{10}} \right) \\ &= 0.3230 \text{ V} \end{aligned}$$

$$\begin{aligned} \therefore x_{dT} &= \sqrt{\frac{4\epsilon_{Si}\phi_F}{qN_A}} = \sqrt{\frac{4(1.036 \times 10^{-12})(0.3230)}{(1.602 \times 10^{-19})(4 \times 10^{15})}} \\ &= 4.570 \times 10^{-5} \text{ cm} \\ &= 0.4570 \mu\text{m} \end{aligned}$$

$$\begin{aligned} |Q_{D,max}| &= q N_A x_{dT} = (1.602 \times 10^{-19})(4 \times 10^{15})(4.570 \times 10^{-5}) \\ &= 2.928 \times 10^{-8} \text{ C/cm}^2 \end{aligned}$$

$$\therefore V_{Th} = V_{FB} + 2\phi_F + \frac{|Q_{D,max}|}{C_{ox}} = -0.9 + 2(0.3230) + \frac{2.928 \times 10^{-8}}{3.453 \times 10^{-7}}$$

$$\therefore V_{GS2} = V_{Th} + 1 = -0.1693 + 1 = 0.8307 \text{ V}$$

Case III: (10x Lower Doping)

$$N_A = 4 \times 10^{13} \text{ cm}^{-3}$$

$$\therefore \phi_F = (0.0259) \ln \left(\frac{4 \times 10^{13}}{1.5 \times 10^{10}} \right)$$

$$= 0.2039 \text{ V}$$

$$\therefore x_{dT} = \sqrt{\frac{4 \epsilon_{Si} \phi_F}{q N_A}} = \sqrt{\frac{4 (1.036 \times 10^{-12}) (0.2039)}{(1.602 \times 10^{-19}) (4 \times 10^{13})}}$$

$$= 3.631 \times 10^{-4} \text{ cm}$$

$$= 3.631 \mu\text{m}$$

$$\therefore |Q_{D, \max}| = q N_A x_{dT} = (1.602 \times 10^{-19}) / (4 \times 10^{13}) (3.631 \times 10^{-4})$$

$$= 2.327 \times 10^{-9} \text{ C/cm}^2$$

$$\therefore V_{Th} = V_{FB} + 2\phi_F + \frac{|Q_{D, \max}|}{C_{ox}}$$

$$= -0.9 + 2(0.2039) + \frac{2.327 \times 10^{-9}}{3.453 \times 10^{-7}}$$

$$= -0.4854 \text{ V}$$

$$\therefore V_{GS} = V_{Th} + 1 = -0.4854 + 1 = 0.5146 \text{ V}$$

$$\therefore N_A \uparrow, x_{dT} \downarrow$$

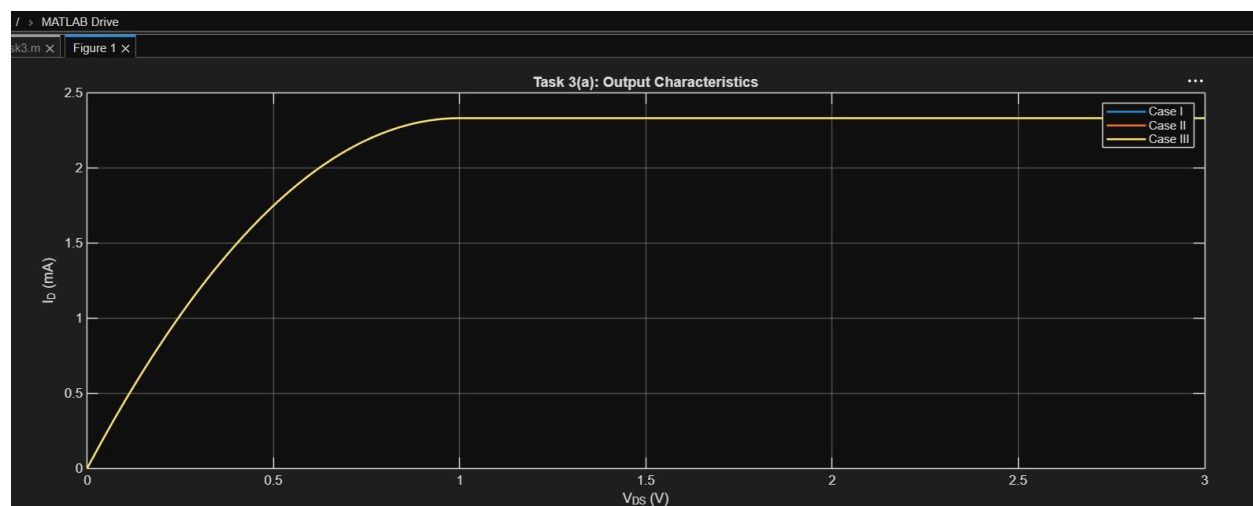
$$\therefore N_A \uparrow, V_{Th} \uparrow$$

```
problem_0001task3.m × Figure 1 × +
/MATLAB Drive/problem_0001task3.m

1 %% Task 3(a)
2 clear; clc; close all;
3
4 q = 1.602e-19;
5 eps0 = 8.854e-14;
6 eps_si = 11.7*eps0;
7 eps_ox = 3.9*eps0;
8
9 tox = 1e-6;
10 Cox = eps_ox/tox;
11
12 W = 10e-4;
13 L = 1e-4;
14 mu = 1350;
15 VFB = -0.9;
16 ni = 1.5e10;
17 kTq = 0.02585;|
18
19 NA = [4e14 4e15 4e13];
20 VDS = linspace(0,3,300);
21
22 for i = 1:3
23
24     phi = kTq*log(NA(i)/ni);
25     Xd = sqrt(4*eps_si*phi/(q*NA(i)));
26     Qd = q*NA(i)*Xd;
```

```
problem_0001task3.m × Figure 1 × +
MATLAB Drive/problem_0001task3.m

28     Vth = VFB + 2*phi + Qd/Cox;
29     Vgs = Vth + 1;
30
31     Vov = Vgs - Vth;
32
33     ID = zeros(size(VDS));
34
35     for j = 1:length(VDS)
36         if VDS(j) < Vov
37             ID(j) = mu*Cox*(W/L) * ...
38                 (Vov*VDS(j) - VDS(j)^2/2);
39         else
40             ID(j) = 0.5*mu*Cox*(W/L)*Vov^2;
41         end
42     end
43
44     plot(VDS,ID*1000,'LineWidth',1.5);
45     hold on;
46
47 end
48
49 xlabel('V_{DS} (V)');
50 ylabel('I_D (mA)');
51 title('Task 3(a): Output Characteristics');
52 legend('Case I','Case II','Case III');
53 grid on;
```



(b) Given that,
 $\beta = 4$

Case I: $N_A = 4 \times 10^{14} \text{ cm}^{-3}$

Case II: $N_A = 4 \times 10^{15} \text{ cm}^{-3}$

Case III: $N_A = 4 \times 10^{13} \text{ cm}^{-3}$

$t_{ox} = 10 \text{ nm}$

$\epsilon_{ox} = 3.9 \epsilon_0$

$L = 1 \mu\text{m}$

$W = 10 \mu\text{m}$

$V_{FB} = -0.9 \text{ V}$

$n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$

$\mu_n = 1350 \text{ cm}^2/\text{Vs}$

$T = 300 \text{ K}$

For Task 3:

$V_{DS} = 0 \rightarrow 3 \text{ V}$

$V_{GS} = V_{Th} + 1 \text{ V}$

For the linear region:

$$I_D = \mu_n C_{ox} \frac{W}{L} \left[(V_{GS} - V_{Th}) V_{DS} - \frac{V_{DS}^2}{2} \right]$$

For saturation,

$$\begin{aligned} I_{Dsat} &= \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{Th})^2 \\ &= \frac{1}{2} \times 1350 \times 3.453 \times 10^{-7} \times \frac{1 \times 10^{-3}}{1 \times 10^{-4}} (1)^2 \\ &= 2.331 \times 10^{-3} \text{ A} \\ &= 2.331 \text{ mA} \end{aligned} \quad \left| \begin{array}{l} V_{GS} = V_{Th} + 1 \\ \Rightarrow V_{GS} - V_{Th} = 1 \text{ V} \\ \Rightarrow V_{DS, sat} = 1 \text{ V} \end{array} \right.$$

$I_{Dsat} = 2.331 \text{ mA}$

The saturation current and operating-region boundary remain almost same because, $V_{GS} - V_{Th} = 1 \text{ V}$ for all cases.

From "a",

Case I:

$N_A = 4 \times 10^{14} \text{ cm}^{-3}$

$\phi_F = 0.2634 \text{ V}$

$x_{dT} = 1.305 \mu\text{m}$

$V_{Th} = -0.3490 \text{ V}$

$V_{GS} = 0.6510 \text{ V}$

Case II:

$N_A = 4 \times 10^{15} \text{ cm}^{-3}$

$\phi_F = 0.3230 \text{ V}$

$x_{dT} = 0.457 \mu\text{m}$

$V_{Th} = -0.1693 \text{ V}$

$V_{GS} = 0.8307 \text{ V}$

Case III:

$N_A = 4 \times 10^{13} \text{ cm}^{-3}$

$\phi_F = 0.2039 \text{ V}$

$x_{dT} = 3.631 \mu\text{m}$

$V_{Th} = -0.4854 \text{ V}$

$V_{GS} = 0.5146 \text{ V}$

$V_{Th} \rightarrow \text{Case III} < \text{Case I} < \text{Case II}$

```
problem_0002task3.m × Figure 1 × +
ATLAB Drive/problem_0002task3.m

1 %% Task 3(b)
2 clear; clc; close all;
3
4 q = 1.602e-19;
5 eps0 = 8.854e-14;
6 eps_si = 11.7*eps0;
7 eps_ox = 3.9*eps0;
8
9 tox = 1e-6;
0 Cox = eps_ox/tox;
1
2 W = 10e-4;
3 L = 1e-4;
4 mu = 1350;
5 VFB = -0.9;
6 ni = 1.5e10;
7 kTq = 0.02585;
8
```

```
problem_0002task3.m × Figure 1 × +
/MATLAB Drive/problem_0002task3.m

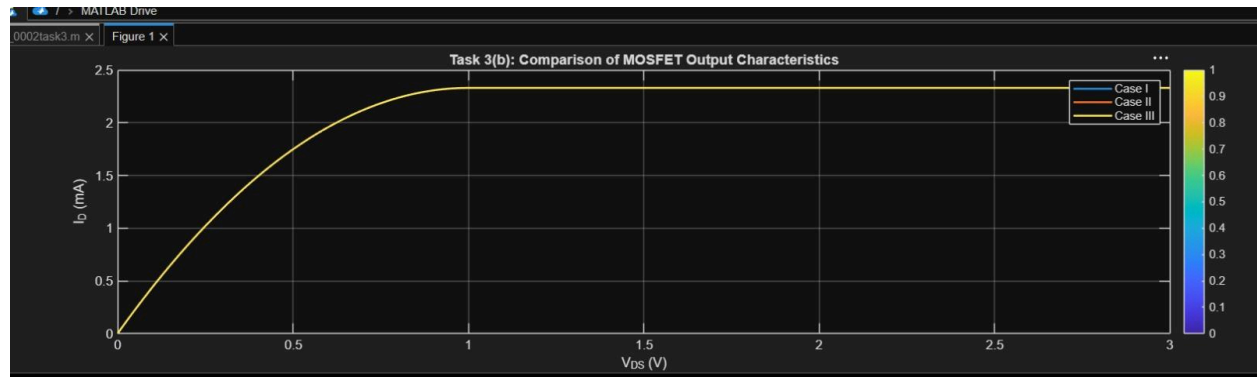
19 NA = [4e14 4e15 4e13];
20 VDS = linspace(0,3,300);
21
22 for i = 1:3
23
24     phi = kTq*log(NA(i)/ni);
25     Xd = sqrt(4*eps_si*phi/(q*NA(i)));
26     Qd = q*NA(i)*Xd;
27
28     Vth = VFB + 2*phi + Qd/Cox;
29     Vgs = Vth + 1;
30
31     Vov = Vgs - Vth;
32
33     ID = zeros(size(VDS));
34
35     for j = 1:length(VDS)
36
```

```
problem_0002task3.m × Figure 1 × +
/MATLAB Drive/problem_0002task3.m
35     for j = 1:length(VDS)
36
37         if VDS(j) < Vov
38             ID(j) = mu*Cox*(W/L) * ...
39                 (Vov*VDS(j) - VDS(j)^2/2);
40         else
41             ID(j) = 0.5*mu*Cox*(W/L)*Vov^2;
42         end
43
44     end
45
46     fprintf('Case %d: Vth = %.4f V, Vgs = %.4f V, Idsat = %.3f mA\n', ...
47         i,Vth,Vgs,max(ID)*1000);
48
49     plot(VDS,ID*1000,'LineWidth',1.5);
50     hold on;
51
52 end
```

```
problem_0002task3.m × Figure 1 × +
/MATLAB Drive/problem_0002task3.m
50     hold on;
51
52 end
53
54 xlabel('V_{DS} (V)');
55 ylabel('I_D (mA)');
56 title('Task 3(b): Comparison of MOSFET Output Characteristics');
57 legend('Case I','Case II','Case III');
58 grid on;
```

Command Window

```
Case 1:  $V_{th} = -0.3489$  V,  $V_{gs} = 0.6511$  V,  $I_{dsat} = 2.331$  mA  
Case 2:  $V_{th} = -0.1693$  V,  $V_{gs} = 0.8307$  V,  $I_{dsat} = 2.331$  mA  
Case 3:  $V_{th} = -0.4854$  V,  $V_{gs} = 0.5146$  V,  $I_{dsat} = 2.331$  mA  
>>
```



(c)

Threshold voltage is the minimum gate-to-source voltage required to form a strong inversion channel between the source and drain of a MOSFET.

For an n-channel MOSFET, the threshold voltage is

$$V_{th} = V_{FB} + 2\phi_F + \frac{|Q_{D,max}|}{C_{ox}}$$

where,

$$\phi_F = \frac{KT}{q} \ln\left(\frac{N_A}{n_i}\right)$$

and,

$$|Q_{D,max}| = \sqrt{4q\epsilon_s N_A \phi_F}$$

$$\therefore N_A \uparrow, \phi_F \uparrow, |Q_{D,max}| \uparrow, V_{th} \uparrow$$

When substrate doping concentration N_A increases, ϕ_F increases because it depends on $\ln(N_A/n_i)$.

The depletion charge $|Q_{D,max}|$ also increases. Therefore, a larger gate voltage is required to create the inversion layer.

$$\therefore N_A \uparrow, V_{th} \uparrow.$$

$$\text{Case III: } N_A = 4 \times 10^{13} \text{ cm}^{-3}, V_{th} = -0.4854 \text{ V}$$

$$\text{Case I: } N_A = 4 \times 10^{14} \text{ cm}^{-3}, V_{th} = -0.3490 \text{ V}$$

$$\text{Case II: } N_A = 4 \times 10^{15} \text{ cm}^{-3}, V_{th} = -0.1693 \text{ V}.$$

Therefore, the result shows that higher substrate doping increases V_{th} .

The channel charge can be approximately expressed as,

$$Q_{inv} \approx C_{ox} (V_{gs} - V_{th})$$

$\therefore V_{gs} \rightarrow \text{fixed}$, substrate doping increases, V_{th} increases,

$$(V_{gs} - V_{th}) \downarrow$$

$$|Q_{inv}| \downarrow$$

So, higher substrate doping reduces the available channel charge at the same gate voltage.

Higher doping means more acceptor atoms are present in the substrate. More charge must be overcome before a strong inversion channel can form. Therefore, a higher gate voltage is needed to produce the same amount of channel charge.

MOSFET transconductance is,

$$g_m = \frac{\partial I_D}{\partial V_{gs}}$$

For the long-channel MOSFET in saturation,

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{gs} - V_{th})^2$$

$$g_m = \mu_n C_{ox} \frac{W}{L} (V_{gs} - V_{th})$$

$\therefore V_{gs} \text{ fix}$, substrate doping increases, $V_{th} \uparrow$
 $(V_{gs} - V_{th}) \downarrow$, $g_m \downarrow$.

$\therefore$ So, higher substrate doping generally reduces transconductance at a fixed V_{gs} .

For saturation, On-current is,

$$I_{Dsat} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{Th})^2$$

$\therefore V_{Th} \uparrow, (V_{GS} - V_{Th}) \downarrow, I_{ON} \downarrow$

$$V_{GS} = V_{Th} + 1$$

$$\Rightarrow V_{GS} - V_{Th} = 1V$$

That is why our simulation gave approximately the same $I_{Dsat} = 2.331 \text{ mA}$ for all three cases.

Implications for digital circuit Design:

a) Noise Margin:

Higher substrate doping increases the threshold voltage.

A higher V_{Th} means the MOSFET needs a larger gate voltage before it turns on strongly. This can help reduce unwanted conduction caused by small voltage disturbances.

Therefore, appropriate substrate doping can help improve the noise margin of a digital circuit.

However, excessively high V_{Th} can make the device slower because more gate voltage is required for strong conduction.

(b) switching speed:

Digital switching speed depends strongly on how quickly the MOSFET can charge and discharge the load.

$\therefore I_{ON} \uparrow$.

$\therefore V_{DS}$ fix, higher substrate doping can reduce the current available to charge or discharge the load.

$$t_{\text{switch}} \propto \frac{CLV}{I_{ON}}$$

$\therefore I_{ON} \downarrow$, $t_{\text{switch}} \uparrow$.

So, higher substrate doping can reduce switching speed when it causes a higher V_{th} and I_{ON} lower.

Conclusion:

Higher substrate doping increases V_{th} and reduces g_m and I_{ON} at fixed V_{DS} . Therefore proper doping is important for balancing noise margin and switching speed.

Task 4

Task 4: Estimating Channel Mobility from Experimental Data

a)

Obtain experimental I–V data by selecting an n-channel MOSFET (e.g., 2N7000 or BS170 or equivalent) from assumption. Assume the drain current (I_D) for at least 10 different gate voltages (V_{G_s}) in the saturation region (keep V_{D_s} constant at 3V or higher). Record V_{th} from your $\sqrt{I_D}$ vs. V_{G_s} plot. Load the data into MATLAB and plot.

b)

Simulate the MOSFET you assumed experimentally using the standard saturation-region model in MATLAB. Use the datasheet values (W , L , t_{ox}) if available, otherwise use the values from your instructor.

- Step 1: Start with $\mu_{eff} = 450 \text{ cm}^2/\text{V}\cdot\text{s}$ and V_{th} extracted from your $\sqrt{I_D}$ plot. Tune μ_{eff} (keeping V_{th} fixed) until the simulated transfer curve resembles the experimental one.
- Step 2: Fix μ_{eff} at the tuned value, then fine-tune V_{th} until the best fit is achieved.

c) Plot both simulated and experimental I_D – V_{G_s} curves (in saturation) on the same graph after optimization. Comment on the goodness of fit.

d) Discuss the physical meaning of the effective channel mobility (μ_{eff}) and explain why the effective mobility in a real MOSFET is lower than the bulk electron mobility. Include effects such as surface scattering, vertical electric field, interface traps, and phonon scattering.

(a)

For an n channel MOSFET (assuming BS170), we assumed our fixed drain to source voltage V_{DS} to be 4V. We varied V_{GS} from 2.3V to 6V (taking 12 data points) and corresponding I_D values were also assumed as specified in the instructions.

V_{GS} (V)	I_D (mA)
2.3	0.0263
2.5	0.1060
2.7	0.2315
3.0	0.5388
3.3	0.9369
3.6	1.494
4.0	2.3378
4.4	3.5122
4.8	4.8146
5.2	6.1602
5.6	8.1364
6.0	9.9614

Next, using the table, we plot V_{GS} vs $\sqrt{I_D}$ in MATLAB with the given code.

```
clear;
clc;
close all;

VGS = [2.3 2.5 2.7 3.0 3.3 3.6 4.0 4.4 4.8 5.2 5.6 6.0];

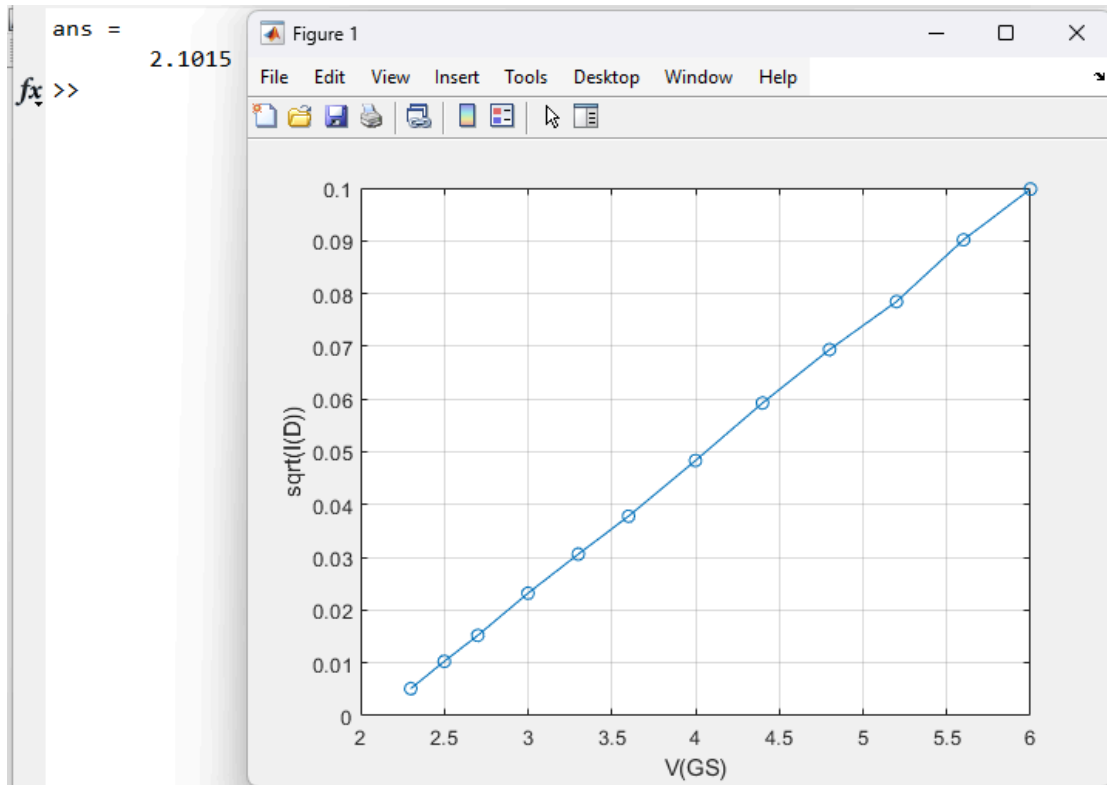
ID = [0.0263 0.1060 0.2315 0.5388 0.9369 1.4294 2.3378 3.5122
4.8146 6.1602 8.1364 9.9614] * 1e-3;

sqrtID = sqrt(ID);

plot(VGS, sqrtID, 'o-');
xlabel('V(GS)');
ylabel('sqrt(I(D))');
grid on;

interp1(sqrtID, VGS, 0, 'linear', 'extrap')
```

Results are



Our taken values go as low as 0.0263 . By using the interpolation function, we find for $I_D = 0$ to get our threshold voltage, V_{TH} . Which we find to be 2.1015 V.

(b)

By plugging the parameters given before, and the coefficient for said MOSFET, $\mu_{\text{eff}} = 450 \text{ cm}^2/\text{V.s}$.

$$I_D = \frac{\mu \cdot C_{ox} \cdot W}{2L} (V_{GS} - V_{TH})^2$$

Using this equation, we now can solve for the actual result through simulating the MOSFET in MATLAB plotting over our expected outcome. The MATLAB code is as follows,

```
clear;
clc;
close all;

VGS = [2.3 2.5 2.7 3.0 3.3 3.6 4.0 4.4 4.8 5.2 5.6 6.0];

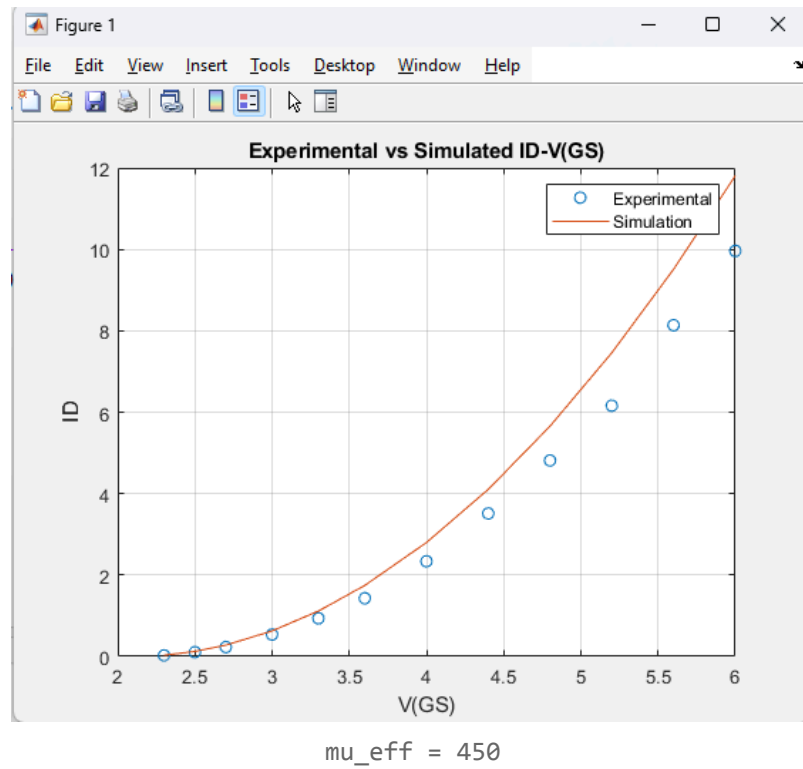
ID = [0.0263 0.1060 0.2315 0.5388 0.9369 1.4294 2.3378 3.5122
4.8146 6.1602 8.1364 9.9614] * 1e-3;

sqrt_ID = sqrt(ID);
VTH = 2.1015;

Cox = (3.9 * 8.854e-14) / (10e-7);
W = 10e-4;
L = 1e-4;
mu_eff = 450;

simID = 0.5 * mu_eff * Cox * (W/L) .* max(VGS - VTH, 0).^2;
plot(VGS, ID*1000, 'o'); hold on;
plot(VGS, simID*1000, '-');
xlabel('V(GS)'); ylabel('ID');
title('Experimental vs Simulated ID-V(GS)');
legend('Experimental', 'Simulation');
grid on;
```

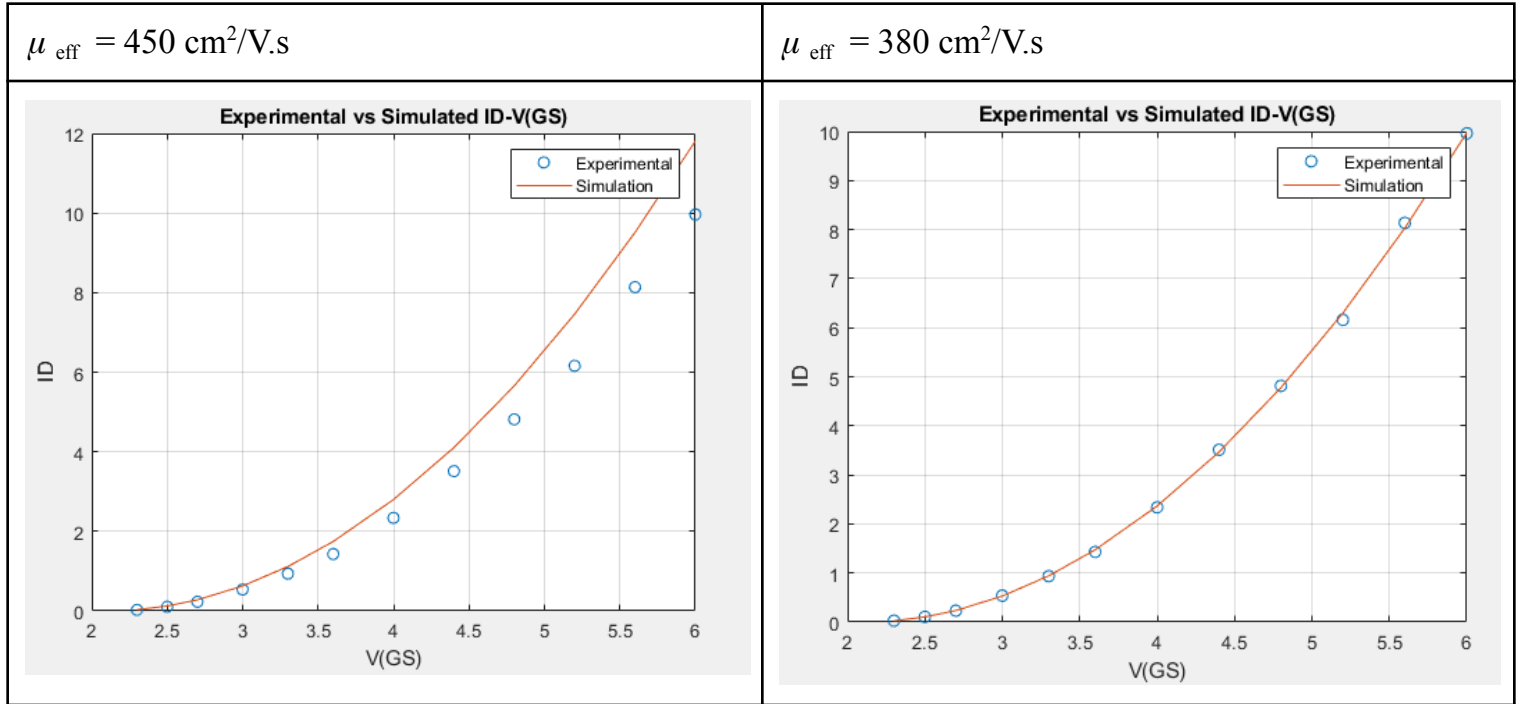
The results were,



As we can see, the simulation didn't match our expected outcome. But after much tuning, changing the value of effective mobility, μ_{eff} to 380 (changing mu_eff to 380; in MATLAB) seemed to have reproduced a simulation result close to our expected outcome.

(c)

Optimizing the simulation by setting $V_{TH} = 2.1015$ V and $\mu_{eff} = 380$ cm²/V.s , we achieved our desired outcome.



After changing, the simulated curve closely follows the experimental data we had taken initially over the V_{GS} range.

(d)

MOSFET channel mobility: In the channel of a MOSFET, charge carriers move from source to drain. Effective channel mobility measures how easily these carriers move under actual operating conditions. Due to several scattering mechanisms, it is lower than the corresponding bulk electron mobility.

Surface scattering: This occurs due to the roughness of the semiconductor-oxide interface. Carriers moving near the surface can be deflected by this roughness. By applying a vertical electric field from the gate towards the interface the carriers are shifted towards the interface, increasing their surface interaction and therefore their scattering.

Interface traps: These are defects at the semiconductor-oxide interface that can interact with charge carriers and cause them to deviate from their normal path.

Phonon scattering: This occurs due to vibrations of atoms in the silicon crystal. These vibrations interfere with the movement of charge carriers and further reduce their mobility.